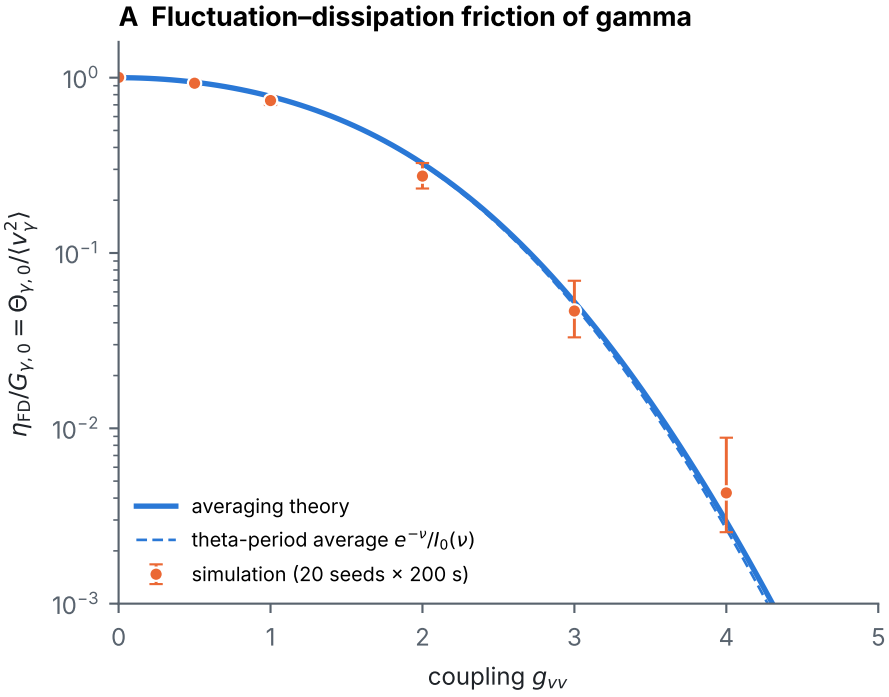
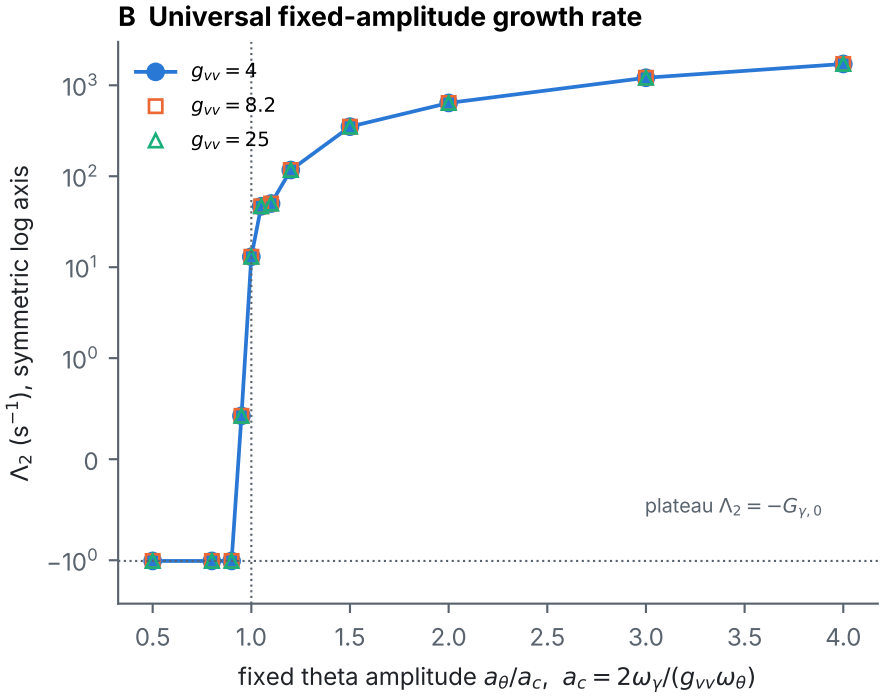

A Fluctuation–dissipation friction of gamma
$\eta_{FD}/G_{\gamma,0} = \Theta_{\gamma,0}/\langle v_\gamma^2 \rangle$
10^0
10^{-1}
10^{-2}
10^{-3}
averaging theory
theta-period average $e^{-\nu}/I_0(\nu)$
simulation (20 seeds × 200 s)
0
1
2
3
4
5
coupling g_{vv}


B Universal fixed-amplitude growth rate
$g_{vv} = 4$
$g_{vv} = 8.2$
$g_{vv} = 25$
Λ_2 (s^{-1}), symmetric log axis
10^3
10^2
10^1
10^0
0
-10^0
plateau $\Lambda_2 = -G_{\gamma,0}$
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
fixed theta amplitude a_θ/a_c, $a_c = 2\omega_\gamma/(g_{vv}\omega_\theta)$